In section 2.3.1, *phonation* was described to the sound production that occurs when air passes through the vocal folds, causing them to vibrate. Based on the categorization of *phonation* by Sundberg into *breathy, pressed, neutral* and *flow*, Proutskova et al., curated the Phonation Mode (Proutskova et al., 2013) dataset. Again, while this dataset is valuable for the classification of phonation modes, its practical application remains somewhat limited when it comes to performance contexts, where phonation modes often change dynamically and are influenced by factors such as phrasing, articulation or vocal dynamics.

2.4 Loudness Estimation Approaches

Loudness to *intensity* is analogous to *pitch* to *fundamental frequency.* While pitch is a perceptual measure, fundamental frequency is the measured frequency from an audio signal. Intensity refers to the measured energy of the signal transmitted per unit of time in the specified direction through a unit area normal to this direction (Fletcher & Munson, 1933), or we can say that intensity is the sound power per unit area. The average threshold of hearing (TOH) or threshold intensity, which can vary by a factor of 10 depending on the listener, is about 2.5×10^{-12} watts per square meter at $1000Hz$ (Backus, 1977). However, practically, instead of using 'watts per square meter' threshold intensity is used a starting point, and correlation is determined in terms of the factor by which a given sound is more intense than the threshold; i.e, if I is the sound intensity and I_0 is the threshold intensity, $\frac{I}{I_0}$. is used as a measure. For a sound of 10^{-8} watts per square meter, this ratio would be 10^4, which is cumbersome number, hence, the intensity is normally measured in the *log* scale given by:

$$L = 10 \log \frac{I}{I_0} \tag{2.1}$$

This gives the level of the threshold of hearing as 0 decibels. Also, increasing the intensity by a factor of 2 translates to changing the intensity level by 3 decibels. Intensity is a physical quantity without any reference to the ear. The microphones usually however measure the pressure amplitude or Sound Pressure level (SPL), which is proportional to the square of the sound pressure. Hence, practically, the following equation is used:

$$L_p = 20 \log \frac{P}{P_0} \tag{2.2}$$

In this equation, P is the *rms* sound pressure level, and L_p is expressed in decibels. Above $1000Hz$, the sensitivity increases with most sensitive regions around the range of $3000 - 4000Hz$, this is partly due to the fact that auditory canal has a resonance frequency in this region (Backus, 1977). Since, the intensity required to hear a sound with similar loudness levels varies with

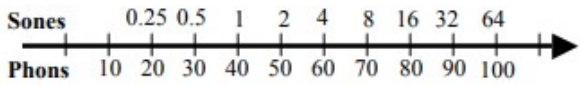

Figure 2.6: Relationship between Sone and Phone

frequency, this provides the foundation of constructing equal-loudness-curves. Empirical studies on measurement of loudness (or perceptual intensity) began as early as 1933 with the publication of equal-loudness-contours (Fletcher & Munson, 1933). A picture of the same is presented in Figure 2.7. The unit of loudness level is named *phon*. One can obtain the loudness level in *phones* given the sound pressure level and frquency from these curves, for example, at $100Hz$, a sound pressure level of $50decibels$ has a loudness of $40phons$. The highest level of $120phons$ is the upper-limit of hearing.

The perceived loudness is a complex phenomenon, determined by several factors, primarily masking and hearing acuity. The human ear is most sensitive to sounds between $2kHz$ and $5kHz$, and less sensitive to higher and lower frequencies. In the human ear, a **critical band** refers to the frequency bandwidth within which the human ear processes sound as a single auditory event or within similar channels. The channels are the critical band filters (Hartmann, 2004). They are not fixed filters but form around the stimulus frequency. When multiple tones fall within the same critical band, they interfere with each other, leading to masking. However, if the tones are outside of the critical band, the ear can distinguish them as separate sounds (Zwicker & Fastl, 2013). Based on the concept of critical bands, Zwicker.... the auditory spectrum is divided into 24 critical bands, reflecting how frequencies are processed. This is measured in the unit 'bark'named after the scientist ' Heinrich Barkhausen' who proposed the first subjective measurement of loudness (Zwicker & Fastl, 2013) While it is beyond the scope of this work to elucidate all concepts concerning loudness but generally loudness is affected by several factors, including but not limited to masking, spectral summing, and temporal effects (Skovenborg & Nielsen, 2004) There are primarily 2 reference standards for computing loudness, (1) Zwicker (2) Moore.

The concept of *sones* was introduced to have a linear mapping of the perception of loudness (Beck & Shaw, 1967). 40 *phons* is arbitrarily set to 1 *sone*. According to the *sone* scale, 2 *sones* is twice as loud as 1 *sone*, 20 *sones* is twice as loud as 10 and so on. Furthermore, loudness measured in the sone scale can be added to find the loudness of the sum of sounds. The results of a number of measurements are represented in 2.6, which relates to loudness in *sones* to loudness level in *phons*. From the Figure, 70 *phons* translates to 8 *sones*, and 80 *phons* to 16 *sones*. The conversion between *sones* and *phons* is

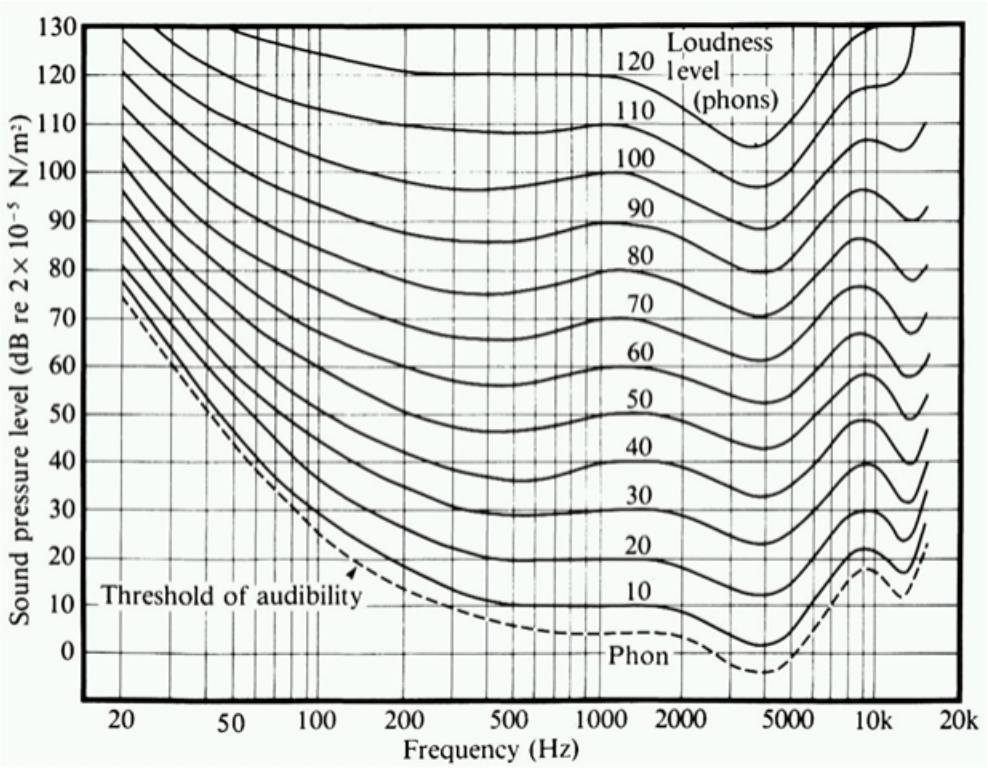

Figure 2.7: Equal Loudness Contours (Phons). Describes the changes in sound pressure level or intensity required to maintain constant loudness for different frequencies. The number on each contour describes the loudness level in phons (Fletcher & Munson, 1933). Figure derived from website: https://learn.microsoft.com/en-us/archive/blogs/audiofool/louder-sounds-better

carried out by the equation:

$$S = \begin{cases} 2^{(L-40)/10}, & \text{if } P >= 40. \\ (L/40)^{2.642}, & P < 40. \end{cases} \tag{2.3}$$

Hence the *sone* scale refers to the perceived loudness.

Several models exist for measuring loudness:

2.4.1 Stevens Power Law

In another model to find the relationship between intensity of a physical stimulus and its perceptual magnitude, Steven employed a linear relationship between the values measures in DB with the values in Sones. It is measured by the equation

$$P = kI^N \tag{2.4}$$

P : Perceived loudness
I : Intensity of the physical stimulus
k : Proportionality constant (varies with stimulus and units)
n : Exponent specific to the type of stimulus

It was met with skepticism because human perception of loudness is inherently nonlinear and depends on various factors, such as frequency, sound pressure levels, and the listener's auditory system.

2.4.2 A-weighting

The A-weihting loudness meter implemented in library corresponds to $40phon$ curve in the Figure. A network labeled B corresponds to $70phon$ curve. Measurements using A-weighting are recorded as dBA. A network usually corresponds well to the subjective impression of the listener to the sound measured.

2.4.3 EBU-R

The EBU R-128 loudness scale is a standard introduced by the European Broadcasting Union (EBU) for measuring and normalizing audio loudness in broadcast and media production. The measurement unit is Loudness Unit Full Scale (LUFS). It is based on ITU-R BS.1770 standard, employing K-weighted filter for loudness computation.

2.5 Related MIR Tasks

To address the task of dynamics analysis, we utilized several existing Music Information Retrieval (MIR) tasks as pre-processing steps, which are illustrated in Figure 2.8. Since the melodic content we analyze is often polyphonic, we apply source separation as a pre-liminary step to get isolated vocals for analysis. This step will be discussed further in Chapter 4.

We also rely on alignment techniques: audio-to-audio alignment for comparing performances, for instance, professional performance with a karaoke performance, and audio-to-score alignment for extracting dynamics annotations and corresponding performance segments. Here, pitch estimation is used as a feature for alignment, while structural segmentation plays a key role in genre-specific analysis, especially for Rock and Pop music. These techniques will be further elaborated in Chapters 4, 5, 6.